Nonlinear Finite Element Methods

Lecture 1: Large Deformation and Reference Formulation

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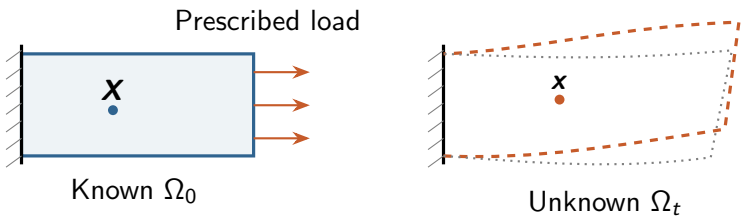
Lecture outline

1. The difficulty: equilibrium on an unknown deformed body.
2. The deformation map and the geometry needed to describe that body.
3. Mapping surface forces and equilibrium to the reference body.
4. The hyperelastic boundary-value problem, variational form, and residual.

The next lecture develops Newton's method for the resulting nonlinear equations.

The loaded body and its unknown shape

The reference body, its supports, and its applied load are given. The displacement determines the current shape.



The dashed and dotted outlines illustrate possible deformed shapes. Which one satisfies equilibrium must be found.

Equilibrium is written on the current body

Let $\boldsymbol{\sigma}$ be Cauchy stress, $\boldsymbol{n}$ the current outward unit normal, ρ current density, and $\boldsymbol{b}$ body force per unit mass. Neglect inertia.

$$-\nabla_{\boldsymbol{x}} \cdot \boldsymbol{\sigma} = \rho \boldsymbol{b} \quad \text{in } \Omega_t.$$

For every current material subdomain ω_t, the equivalent force balance is

$$\boxed{\int_{\partial \omega_t} \boldsymbol{\sigma} \boldsymbol{n} \, \mathrm{d}a + \int_{\omega_t} \rho \boldsymbol{b} \, \mathrm{d}V_t = \mathbf{0}.}$$

The domain, its boundary normals and areas, and the stress depend on the displacement being sought.
We choose a reference formulation so that integration is performed on the known body Ω_0. Current-configuration formulations are also possible.

The deformation map and its gradient

Label a material point by $\boldsymbol{X} \in \Omega_0$. Its current position is

$$\boldsymbol{x} = \boldsymbol{\varphi}(\boldsymbol{X}) = \boldsymbol{X} + \boldsymbol{u}(\boldsymbol{X}).$$

Taking the total differential of $x_i = \varphi_i(\boldsymbol{X})$ gives

$$\mathrm{d}x_i = \frac{\partial \varphi_i}{\partial X_J}\,\mathrm{d}X_J = F_{iJ}\,\mathrm{d}X_J.$$

Define the **deformation gradient**:

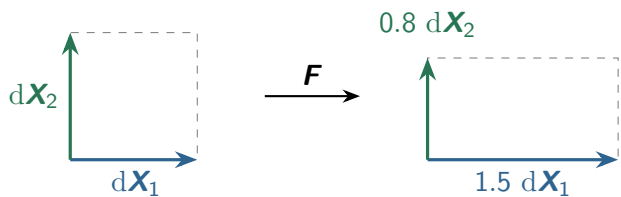
$$\boxed{\boldsymbol{F} = \operatorname{Grad}\boldsymbol{\varphi} = \boldsymbol{I} + \operatorname{Grad}\boldsymbol{u}, \qquad \mathrm{d}\boldsymbol{x} = \boldsymbol{F}\,\mathrm{d}\boldsymbol{X}.}$$

Here Grad means differentiation with respect to $\boldsymbol{X}$; $(\operatorname{Grad}\boldsymbol{u})_{iJ} = \partial u_i/\partial X_J$. Repeated indices are summed.
Assume smooth, one-to-one, orientation-preserving deformations.

Example: local deformation

In a small material region, consider the following local deformation:

$$\boldsymbol{x} = \begin{bmatrix} 1.5X_1 \\ 0.8X_2 \\ X_3 \end{bmatrix}, \qquad \boldsymbol{F} = \begin{bmatrix} 1.5 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$



A line parallel to the first axis lengthens by 50%. A line parallel to the second axis shortens by 20%.

Measuring deformation: $\boldsymbol{C}$ and $\boldsymbol{E}$

The line mapping gives

$$\mathrm{d}\boldsymbol{x}\cdot\,\mathrm{d}\boldsymbol{x} = (\boldsymbol{F}\,\mathrm{d}\boldsymbol{X})\cdot(\boldsymbol{F}\,\mathrm{d}\boldsymbol{X}) = \mathrm{d}\boldsymbol{X}\cdot(\boldsymbol{F}^{T}\boldsymbol{F})\,\mathrm{d}\boldsymbol{X}.$$

Define the **right Cauchy–Green tensor** $\boldsymbol{C} = \boldsymbol{F}^{T}\boldsymbol{F}$. The change in squared length is

$$|\,\mathrm{d}\boldsymbol{x}|^2 - |\,\mathrm{d}\boldsymbol{X}|^2 = \mathrm{d}\boldsymbol{X}\cdot(\boldsymbol{C}-\boldsymbol{I})\,\mathrm{d}\boldsymbol{X}.$$

The **Green–Lagrange strain** is therefore

$$\boxed{\boldsymbol{E} = \tfrac{1}{2}(\boldsymbol{C}-\boldsymbol{I}), \qquad |\,\mathrm{d}\boldsymbol{x}|^2 - |\,\mathrm{d}\boldsymbol{X}|^2 = 2\,\mathrm{d}\boldsymbol{X}\cdot\boldsymbol{E}\,\mathrm{d}\boldsymbol{X}.}$$

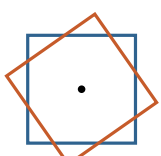
For the preceding example, $C_{11} = 1.5^2 = 2.25$ and $E_{11} = (2.25-1)/2 = 0.625$.

Why infinitesimal strain is insufficient

Substitute $\boldsymbol{F} = \boldsymbol{I} + \operatorname{Grad} \boldsymbol{u}$ into $\boldsymbol{E} = (\boldsymbol{F}^T \boldsymbol{F} - \boldsymbol{I})/2$:

$$\boxed{\boldsymbol{E} = \tfrac{1}{2} \left(\operatorname{Grad} \boldsymbol{u} + (\operatorname{Grad} \boldsymbol{u})^T + (\operatorname{Grad} \boldsymbol{u})^T \operatorname{Grad} \boldsymbol{u} \right).}$$

Infinitesimal strain neglects the last, quadratic term. That approximation requires a small displacement gradient.



A rigid rotation changes orientation but not material lengths. Its deformation gradient satisfies

$$\boldsymbol{F}^T \boldsymbol{F} = \boldsymbol{I}, \qquad \boldsymbol{E} = \boldsymbol{0}.$$

Retaining the quadratic term makes this check hold even for a large rotation.

Volume mapping

Define $J = \det \boldsymbol{F}$. Corresponding infinitesimal volumes satisfy

$$\boxed{\mathrm{d}V_t = J\,\mathrm{d}V_0.}$$

For the same numerical deformation,

$$J = (1.5)(0.8)(1) = 1.2.$$

A reference material region of volume 1 has current volume 1.2. For any integrable current field g,

$$\boxed{\int_{\Omega_t} g(\boldsymbol{x})\,\mathrm{d}V_t = \int_{\Omega_0} g(\boldsymbol{\varphi}(\boldsymbol{X}))J(\boldsymbol{X})\,\mathrm{d}V_0.}$$

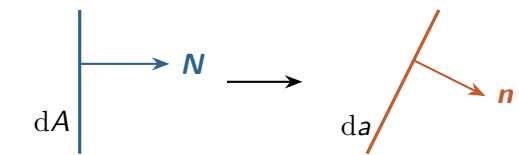
We require $J > 0$ locally. Together with the assumed one-to-one map, this allows the change of variables to the reference body.

Surface forces need an area and a normal mapping

On a current surface patch,

$$\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n}, \qquad \mathrm{d}\boldsymbol{f} = \boldsymbol{\sigma}\boldsymbol{n}\,\mathrm{d}a.$$

Let $\mathrm{d}A$ and $\boldsymbol{N}$ be the corresponding reference area and outward unit normal. To express the same force using them, we must map both the area and its normal.



The surface patches are shown in cross-section. The general map can change their orientation as well as their area.

Nanson's formula: mapping the oriented area

Let two reference edge vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ span a small surface patch:

$$\boldsymbol{N}\,\mathrm{d}A = \boldsymbol{A} \times \boldsymbol{B}.$$

The corresponding current edges are $\boldsymbol{FA}$ and $\boldsymbol{FB}$. Using the cross-product transformation,

$$\begin{aligned}\boldsymbol{n}\,\mathrm{d}a &= (\boldsymbol{FA}) \times (\boldsymbol{FB}) \\ &= J\boldsymbol{F}^{-T}(\boldsymbol{A} \times \boldsymbol{B}) \\ &= J\boldsymbol{F}^{-T}\boldsymbol{N}\,\mathrm{d}A.\end{aligned}$$

Thus **Nanson's formula** is

$$\boxed{\boldsymbol{n}\,\mathrm{d}a = J\boldsymbol{F}^{-T}\boldsymbol{N}\,\mathrm{d}A.}$$

The vector on the right contains both the current normal direction and the current area. $\boldsymbol{F}^{-T}\boldsymbol{N}$ is generally not a unit vector.

First Piola–Kirchhoff stress

Define the **first Piola–Kirchhoff stress $\boldsymbol{P}$** by

$$\mathrm{d}\boldsymbol{f} = \boldsymbol{P}\boldsymbol{N}\,\mathrm{d}A.$$

Insert Nanson's formula into the current force expression:

$$\begin{aligned}\mathrm{d}\boldsymbol{f} &= \boldsymbol{\sigma}\boldsymbol{n}\,\mathrm{d}a \\ &= \boldsymbol{\sigma}\left(J\boldsymbol{F}^{-T}\boldsymbol{N}\,\mathrm{d}A\right) \\ &= \left(J\boldsymbol{\sigma}\boldsymbol{F}^{-T}\right)\boldsymbol{N}\,\mathrm{d}A.\end{aligned}$$

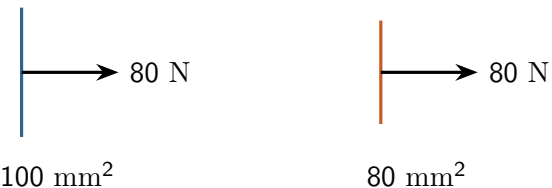
Therefore

$$\boxed{\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-T}, \qquad \boldsymbol{\sigma} = \frac{1}{J}\boldsymbol{P}\boldsymbol{F}^{T}.}$$

$\boldsymbol{P}\boldsymbol{N}$ is physical force per unit reference area. $\boldsymbol{P}$ is generally nonsymmetric, even though $\boldsymbol{\sigma}$ is symmetric.

The same force divided by different areas

For $\boldsymbol{F} = \mathrm{diag}(1.5, 0.8, 1)$, a face normal to the first axis keeps its normal direction, but its area is multiplied by 0.8.



The normal stress values are

$$P_{11} = \frac{80\ \mathrm{N}}{100\ \mathrm{mm}^2} = 0.8\ \mathrm{MPa}, \qquad \sigma_{11} = \frac{80\ \mathrm{N}}{80\ \mathrm{mm}^2} = 1\ \mathrm{MPa}.$$

They agree with $P_{11} = (J/F_{11})\sigma_{11} = (1.2/1.5)\sigma_{11}$. The force is unchanged. Only its area basis differs.

Force balance on the reference body

Return to current equilibrium on $\omega_t = \boldsymbol{\varphi}(\omega_0)$:

$$\mathbf{0} = \int_{\partial\omega_t} \boldsymbol{\sigma}\boldsymbol{n}\,\mathrm{d}a + \int_{\omega_t} \rho\boldsymbol{b}\,\mathrm{d}V_t.$$

Use $\boldsymbol{\sigma}\boldsymbol{n}\,\mathrm{d}a = \boldsymbol{P}\boldsymbol{N}\,\mathrm{d}A$ and $\mathrm{d}V_t = J\,\mathrm{d}V_0$:

$$\mathbf{0} = \int_{\partial\omega_0} \boldsymbol{P}\boldsymbol{N}\,\mathrm{d}A + \int_{\omega_0} \rho(\boldsymbol{\varphi}(\boldsymbol{X}))\boldsymbol{b}(\boldsymbol{\varphi}(\boldsymbol{X}))J\,\mathrm{d}V_0.$$

Define body force per unit reference volume by

$$\boldsymbol{b}_0(\boldsymbol{X}) = J(\boldsymbol{X})\rho(\boldsymbol{\varphi}(\boldsymbol{X}))\boldsymbol{b}(\boldsymbol{\varphi}(\boldsymbol{X})).$$

The same equilibrium statement is now

$$\boxed{\int_{\partial\omega_0} \boldsymbol{P}\boldsymbol{N}\,\mathrm{d}A + \int_{\omega_0} \boldsymbol{b}_0\,\mathrm{d}V_0 = \mathbf{0}.}$$

Current and reference equilibrium

Apply the reference divergence theorem to the preceding surface integral:

$$\int_{\omega_0} (\operatorname{Div} \boldsymbol{P} + \boldsymbol{b}_0)\, \mathrm{d}V_0 = \boldsymbol{0}.$$

For arbitrary ω_0, with sufficiently regular fields, this gives

Current body	Reference body
$-\nabla_{\boldsymbol{x}} \cdot \boldsymbol{\sigma} = \rho \boldsymbol{b}$	$-\operatorname{Div} \boldsymbol{P} = \boldsymbol{b}_0$
$\boldsymbol{x} \in \Omega_t$	$\boldsymbol{X} \in \Omega_0$

The operators use different coordinates:

$$(\nabla_{\boldsymbol{x}} \cdot \boldsymbol{\sigma})_i = \frac{\partial \sigma_{ij}}{\partial x_j}, \qquad (\operatorname{Div} \boldsymbol{P})_i = \frac{\partial P_{iJ}}{\partial X_J}.$$

These are descriptions of the same force balance, not different physical models.

Boundary data must use the corresponding area

Partition the reference boundary into Γ_{0u} and Γ_{0t}:

$$\boldsymbol{u} = \overline{\boldsymbol{u}} \quad \text{on } \Gamma_{0u}, \qquad \boldsymbol{P}\boldsymbol{N} = \overline{\boldsymbol{t}}_0 \quad \text{on } \Gamma_{0t}.$$

On the corresponding current traction boundary $\Gamma_{tt} = \boldsymbol{\varphi}(\Gamma_{0t})$,

$$\boldsymbol{\sigma}\boldsymbol{n} = \overline{\boldsymbol{t}}.$$

The prescribed tractions represent the same force when

$$\boxed{\overline{\boldsymbol{t}}_0 \, \mathrm{d}A = \overline{\boldsymbol{t}} \, \mathrm{d}a.}$$

For our loaded body, prescribe both $\overline{\boldsymbol{t}}_0$ per reference area and $\boldsymbol{b}_0$ per reference volume independently of displacement.
Holding a traction per reference area fixed is generally different from holding its current-area value fixed while the surface changes.

Hyperelastic stress law

A hyperelastic material supplies stored energy per reference volume, $\Psi(\boldsymbol{F})$, and the stress law

$$\boldsymbol{P} = \frac{\partial \Psi}{\partial \boldsymbol{F}}.$$

Recall the compressible neo-Hookean model from Chapter 4:

$$\Psi(\boldsymbol{F}) = \frac{\mu}{2}(\operatorname{tr}(\boldsymbol{F}^T\boldsymbol{F}) - 3) - \mu \ln J + \frac{\lambda}{2}(\ln J)^2, \qquad J > 0.$$

Its first Piola–Kirchhoff stress is

$$\boxed{\boldsymbol{P}(\boldsymbol{F}) = \mu(\boldsymbol{F} - \boldsymbol{F}^{-T}) + \lambda \ln J\, \boldsymbol{F}^{-T}.}$$

Here μ and λ are the material's Lamé constants. The stress depends on $\boldsymbol{u}$ through $\boldsymbol{F} = \boldsymbol{I} + \operatorname{Grad} \boldsymbol{u}$, including its inverse and determinant.

Reference-configuration strong problem

With $\boldsymbol{F}(\boldsymbol{u}) = \boldsymbol{I} + \operatorname{Grad} \boldsymbol{u}$, the problem is

$$\begin{cases} -\operatorname{Div} \boldsymbol{P}(\boldsymbol{F}(\boldsymbol{u})) = \boldsymbol{b}_0 & \text{in } \Omega_0, \\ \boldsymbol{u} = \overline{\boldsymbol{u}} & \text{on } \Gamma_{0u}, \\ \boldsymbol{P}(\boldsymbol{F}(\boldsymbol{u}))\boldsymbol{N} = \overline{\boldsymbol{t}}_0 & \text{on } \Gamma_{0t}. \end{cases} \qquad \text{Strong form}$$

The reference mesh, integration domain, and material labels are known.
The deformation and stress remain unknown. For the neo-Hookean law, $\boldsymbol{F}^{-T}$ and $\ln(\det \boldsymbol{F})$ depend nonlinearly on $\operatorname{Grad} \boldsymbol{u}$.
A guessed displacement lets us evaluate the stress, but that stress generally does not satisfy equilibrium. The displacement must be corrected.

Trial and test functions on the known reference body

Use the Chapter 4 trial and test spaces:

$$\mathcal{U}_{\overline{\boldsymbol{u}}} = \{\boldsymbol{w} \in H^1(\Omega_0; \mathbb{R}^3) : \boldsymbol{w} = \overline{\boldsymbol{u}} \text{ on } \Gamma_{0u}\},$$
$$\mathcal{V}_0 = \{\boldsymbol{v} \in H^1(\Omega_0; \mathbb{R}^3) : \boldsymbol{v} = \boldsymbol{0} \text{ on } \Gamma_{0u}\}.$$

Assume admissible deformations with $J > 0$ and enough regularity for the following stress integrals to be finite. For $\boldsymbol{v} \in \mathcal{V}_0$, start with $-\operatorname{Div} \boldsymbol{P} = \boldsymbol{b}_0$:

$$-\int_{\Omega_0} (\operatorname{Div} \boldsymbol{P}) \cdot \boldsymbol{v} \, \mathrm{d}V_0 = \int_{\Omega_0} \boldsymbol{b}_0 \cdot \boldsymbol{v} \, \mathrm{d}V_0.$$

The unknown and test functions are defined using the fixed coordinates $\boldsymbol{X}$.

Reference-domain variational problem

Integration by parts in the preceding equation gives

$$\int_{\Omega_0} \boldsymbol{P} : \operatorname{Grad} \boldsymbol{v} \, \mathrm{d}V_0 - \int_{\partial\Omega_0} (\boldsymbol{PN}) \cdot \boldsymbol{v} \, \mathrm{d}A = \int_{\Omega_0} \boldsymbol{b}_0 \cdot \boldsymbol{v} \, \mathrm{d}V_0.$$

On Γ_{0u}, $\boldsymbol{v} = \boldsymbol{0}$. On Γ_{0t}, $\boldsymbol{PN} = \overline{\boldsymbol{t}}_0$. Substituting these conditions gives

$$\int_{\Omega_0} \boldsymbol{P}(\boldsymbol{F}(\boldsymbol{u})) : \operatorname{Grad} \boldsymbol{v} \, \mathrm{d}V_0 = \int_{\Omega_0} \boldsymbol{b}_0 \cdot \boldsymbol{v} \, \mathrm{d}V_0 + \int_{\Gamma_{0t}} \overline{\boldsymbol{t}}_0 \cdot \boldsymbol{v} \, \mathrm{d}A.$$

Find $\boldsymbol{u} \in \mathcal{U}_{\overline{\boldsymbol{u}}}$ such that this holds for every $\boldsymbol{v} \in \mathcal{V}_0$.
The gradient is $\operatorname{Grad} \boldsymbol{v}$, not its symmetric part, because $\boldsymbol{P}$ is generally nonsymmetric.

Why the finite element equations are nonlinear

Write the internal form and external load functional as

$$a_{\mathrm{hyp}}(\boldsymbol{u};\boldsymbol{v}) = \int_{\Omega_0} \boldsymbol{P}(\boldsymbol{F}(\boldsymbol{u})) : \operatorname{Grad} \boldsymbol{v}\, \mathrm{d}V_0,$$

$$\ell_{\mathrm{hyp}}(\boldsymbol{v}) = \int_{\Omega_0} \boldsymbol{b}_0 \cdot \boldsymbol{v}\, \mathrm{d}V_0 + \int_{\Gamma_{0t}} \overline{\boldsymbol{t}}_0 \cdot \boldsymbol{v}\, \mathrm{d}A.$$

The residual is

$$\boxed{R(\boldsymbol{u};\boldsymbol{v}) = \ell_{\mathrm{hyp}}(\boldsymbol{v}) - a_{\mathrm{hyp}}(\boldsymbol{u};\boldsymbol{v}).}$$

Equilibrium requires $R(\boldsymbol{u};\boldsymbol{v}) = 0$ for every test function.
After FE approximation, the stress depends on the unknown displacement coefficients. We must solve nonlinear equations for those coefficients, rather than one system with a fixed stiffness matrix.

Summary and next step

- Large deformation changes the body on which equilibrium is written. We choose a known reference domain for the FE formulation.
- The deformation gradient maps line elements. Its determinant maps volume. Nanson's formula maps oriented surface area.
- Preserving physical force gives $\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-T}$ and reference equilibrium $-\operatorname{Div}\boldsymbol{P} = \boldsymbol{b}_0$.
- Mapping fixes the integration domain, but does not make $\boldsymbol{P}(\boldsymbol{I} + \operatorname{Grad}\boldsymbol{u})$ linear in displacement.

The next step is to linearize the residual and assemble the Newton correction system for the hyperelastic FE problem.

Reading and stress convention

- Course book, Chapter 4, *Finite deformation and hyperelasticity*: the material model and reference variational problem used here.
- Local Belytschko notes, Chapter 2, §2.2: material coordinates and the reference-area description of force.
- Local Belytschko notes, Chapter 3, §3.2–3.4: deformation gradient, Green strain, and surface/stress transformations. Nanson's relation appears in §3.4.2, printed p. 3-33.

Our convention is $\boldsymbol{t}_0 = \boldsymbol{P}\boldsymbol{N}$ and $\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-T}$. The local notes' nominal stress uses the transpose convention. All formulas in these slides follow the course book.